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PERIOPERATIVE

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Perioperative Management of Steroids

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Key Points

- All glucocorticoid-dependent patients are at risk of adrenal crisis as a consequence of surgical stress or illness
- Peri-operative steroid supplementation is *in addition* to baseline steroid therapy which should continue throughout the peri-operative period.

Introduction

- All glucocorticoid-dependent patients are at risk of adrenal crisis as a consequence of surgical stress or illness.
    - Primary (adrenal gland disease) and secondary adrenal insufficiency (failure of regulatory centres)
    - Patients on glucocortocoids for other medical conditions: Assume at risk of adrenal insufficiency if receiving a course of prednisolone (or equivalent) of ≥5mg for at least 4 weeks. (Table 1 for prednisolone equivalency).
- If in doubt, give supplemental steroids.
- Dexamethasone is not adequate as glucocorticoidtreatment in patients with primary adrenal insufficiency as it has no mineralocorticoid activity.

	Equivalent Dose (mg)	Glucocorticoid activity	Mineralocorticoid activity
Hydrocortisone	40	1	1
Prednisolone	10	4	0.8
Methylprednisolone	10	5	0.5
Dexamethasone	1.5	25	0
Beclomethasone (inh)	1.5	*Adult asthma dose; 200-400mcg BD, up to 800mcg BD*	
Fluticasone (inh)	0.75	*Adult asthma dose; 100-500mcg BD, up to 1mg BD*	

Table 1

Peri-Operative Considerations

- Establish steroid history, then follow guidance below.
- Post-operative hydrocortisone doses given by infusion, create more physiological glucocorticoid levels, and are preferred to boluses.
- If the patient is still receiving steroid therapy, this should be continued the morning of surgery and through the peri-operative period. If Nil By Mouth, this should be converted to a parenteral form *in addition* to the steroid supplementation.

Adults on ≥5mg Prednisolone (or equivalent) for at least 4 weeks		
Type of Surgery	**Intra-operative replacement**	**Postoperative replacement**
Body surface & Intermediate	Dexamethasone 6-8mg iv	Double regular glucocorticoid dose for 48hrs
Major Surgery including Caesarean section	Dexamethasone 6-8mg iv	Hydrocortisone 4mg/hr (100mg in 24hrs) iv infusion whilst nil by mouth, then double oral dose for up to 7 days unless recovery is uncomplicated.
Labour & Vaginal delivery	Hydrocortisone 100mg iv at onset of labour, followed by immediate initiation of hydrocortisone 4mg/hr. Or Hydrocortisone 100mg im followed by 50mg every 6hrs im	

Primary & Secondary Adrenal Insufficiency		
Type of Surgery	**Intra-operative replacement**	**Postoperative replacement**
Surgery under anaesthesia (GA or RA)	Hydrocortisone 100mg iv on induction, then immediate initiation of 8mg/hr (200mg in 24hrs)	Hydrocortisone 8mg/hr iv infusion while nil by mouth or if postoperative vomiting. When resume oral-double hydrocortisone doses for 48hrs or for up to a week following major surgery
Bowel procedures requiring laxatives/enema	Consider iv fluids and hydrocortisone iv, especially for fludrocortsone or vasopressin-dependant patients. Hydrocortisone 100mg iv/im at start of procedure.	Resume oral- Double hydrocortisone doses for 24hrs
Labour & Vaginal delivery	Hydrocortisone 100mg iv at onset of labour, followed by immediate initiation of hydrocortisone 8mg/hr. Or Hydrocortisone 100mg im followed by 50mg every 6hrs im	Resume oral-Double hydrocortisone doses for 24hrs

Children with Adrenal Insufficiency		
Type of Surgery	**Intra-operative replacement**	**Postoperative replacement**
Major Surgery under anaesthesia (GA or RA)	Hydrocortisone 2mg/kg at induction., then immediate iv infusion: ≤10KG:25mg/24hrs 11-20kg:50mg/24hrs >20kg Prepubertal: 100mg/24hrs >20kg pubertal:150mg/24hrs.	Hydrocortisone 2mg/kg 4hrly iv or im. Once stable, double usual oral hydrocortisone doses for 48hrs, then reduce to normal doses over up to a week.
Minor procedures requiring general anaesthesia	Hydorcortisone 2mg/kg iv on induction	Double usual oral hydrocortisone doses for 24hrs
Minor procedures NOT requiring GA	Double morning dose of hydrocortisone pre-operatively	Resume usual dose of hydrocortisone

References

Woodcock et al. Perioperative management of glucocortocoids. Anaesthesia 2020.